

2. Day of the Week

Given a date, return the corresponding day of the week for that date.

The input is given as three integers representing the day, month and year respectively.

Return the answer as one of the following values {"Sunday", "Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday"}.

Note: January 1, 1971 was a Friday.

Example 1:

Input: day = 31, month = 8, year = 2019

Output: "Saturday"

Example 2:

Input: day = 18, month = 7, year = 1999

Output: "Sunday"

Example 3:

Input: day = 15, month = 8, year = 1993

Output: "Sunday"

Constraints:

- The given dates are valid dates between the years 1971 and 2100.

PROGRAM

```
class Solution {  
  
    public String dayOfTheWeek(int day, int month, int year) {  
  
        String[] days = {  
  
            "Sunday", "Monday", "Tuesday", "Wednesday",  
  
            "Thursday", "Friday", "Saturday"  
  
        };  
  
  
        int totalDays = 0;  
  
  
        // Count days in previous years  
        for (int y = 1971; y < year; y++) {  
  
            totalDays += isLeapYear(y) ? 366 : 365;  
  
        }  
    }  
}
```

```
int[] monthDays = {  
    31, 28, 31, 30, 31, 30,  
    31, 31, 30, 31, 30, 31  
};  
  
// Count days in previous months  
for (int m = 1; m < month; m++) {  
    totalDays += monthDays[m - 1];  
}  
  
// Add extra day for February in leap year  
if (month > 2 && isLeapYear(year)) {  
    totalDays++;  
}  
  
// Add current day - 1  
totalDays += day - 1;  
  
// January 1, 1971 = Friday  
// Friday has index 5  
return days[(totalDays + 5) % 7];  
}  
  
private boolean isLeapYear(int year) {  
    return year % 400 == 0 ||
```

}

